

Lucas Picard

B.S./M.S. Mechanical + B.S. Robotics Engineering | Mechatronics · CAD · Embedded Systems · Prototyping

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Portfolio

SUMMARY

Mechanical engineering student at WPI (3.75 GPA, Dean's List) on a combined B.S./M.S. in Mechanical Engineering with a second major in Robotics Engineering. I design and build real hardware, including embedded electronics, robotic systems, and CAD assemblies, and I direct AI to write the software and automation around them. Dual US/German citizen, bilingual in English and German.

EXPERIENCE

Microboard Processing - *Engineering Intern, Document & Controls*
Seymour, CT - contract electronics manufacturer (ITAR-regulated).

Jun - Aug 2026

- Automated documentation work that engineering had been doing by hand, alongside an industrial engineer.
- Wrote the standard process-flow documents for SMT and PCB assembly, so every build follows the same written steps.
- Handled ITAR-controlled documentation to the company rules.

Picard Studio - *Studio Assistant & Operations*

Feb 2020 - Present

- Built and maintained KPI dashboards tracking MRR, ARPU, retention, churn, and acquisition, which the business uses to make decisions.
- Automated monthly metrics reporting through the Kajabi API and GitHub Actions, sending data straight into Google Sheets.
- Delivered freelance CAD work - designing, prototyping, and invoicing a wall-mount equipment cradle.

EDUCATION

Worcester Polytechnic Institute - *B.S./M.S. Mechanical Engineering + B.S. Robotics Engineering*

Expected 2029

Worcester, MA - 21 AP transfer credits. GPA: 3.75 / 4.00. Dean's List (Fall 2025, Spring 2026).

Completed coursework: Statics, Dynamics, Thermodynamics, Manufacturing Science & CNC Machining, Intro to Robotics, Intro to ECE, Calculus III-IV, Differential Equations, Linear Algebra.

WPI activities: Combat Robotics, Club Badminton, Intramural Soccer.

Pomperaug High School - *Southbury, CT*

2021 - 2025

ENGINEERING PROJECTS

SCARA Robot Arm - *Personal mechatronics build*

2026 - Present

- Turned a salvaged \$150 Anet A8 3D printer into a teleoperated 4-axis SCARA arm for about \$120 in new parts, with my own 3D-printed arm segments and GT2 pulleys up to 100 teeth.
- An Xbox controller drives the arm at 25 Hz, with a Python host solving the inverse kinematics live and sending G-code to Marlin.
- Integrated AS5600 absolute encoders on a three-bus I²C design for switch-free homing; base, elbow, and Z are live, with wrist and gripper next.

PenGPT AI Smart Pen - *Embedded hardware / PCB design*

Mar 2026 - Present

- Co-designing the hardware with a teammate for a pen that turns handwriting on ordinary paper into text, tracking its own motion with two IMUs and a magnetometer around a Sophgo SG2002 Linux processor. Boards are targeted for November 2026.
- Verified the 268-part KiCad schematic pin by pin against manufacturer datasheets, catching three faults ERC missed. The 4-layer bring-up board is placed and routing is next.

CT Used-Car Scraper - *Python automation tool*

2026

- Directed AI to build a Python + Playwright system that scrapes 36 Connecticut dealer sites every 6 hours on GitHub Actions for \$0/month, tracks price history in SQLite, scores deals, and sends Discord alerts. It found the car I bought.

FRC Robotics - *Team Captain & Lead Driver · 1000+ hours*

2022 - 2025

- Captain for two of three seasons. Set the team's direction and worked with the subteam leads, including marketing and outreach. Built a preseason system of lessons and assignments to train new students, then moved the team onto project-management software to assign roles and tasks.
- Introduced CAD to a team that had none and was its only CAD student, taking the robot from taped-together scrap to fully modeled designs, and ran four iterations with subsystem prototypes in one season, up from a single un-prototyped design.
- Replaced the team's WAGO wiring with JST and bullet connectors, soldering every motor and controller over one February vacation, which ended the disconnections that had been costing matches.
- Started the team's presentations to the district Board of Education, schools and community groups. Worked 60-80 hour weeks through build season alongside a full course load, and as primary driver earned the team's first New England District Championship qualification in 10 years and its first CT State Championship.

TECHNICAL SKILLS

CAD & Prototyping: SolidWorks, Onshape, GD&T, design for manufacturability, 3D printing, manual milling, laser cutting, CAM

Electronics: KiCad schematic & PCB design, soldering, embedded systems (ESP32 / Arduino), closed-loop control, sensors & encoders

Software & Data: Java (semi-proficient), read Python, and direct AI to build and ship the rest: automation scripts, web scrapers, REST API integrations, GitHub Actions, and robot control hosts. Technical documentation, ITAR-aware data handling.

AWARDS & LANGUAGES

Awards: Dean's List (WPI, Fall 2025 and Spring 2026); Honor Roll (Pomperaug High School, all years).

Languages: English (native), German (Seal of Bilingualism, dual US/German citizen).